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Q-Trust AI: Hybrid Architecture for Artificial Image Detection

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Abstract

Q-Trust AI is a hybrid quantum–classical architecture for detecting AI-generated images. It combines semantic features from a partially fine-tuned ResNet18 with spectral descriptors obtained through the Fast Fourier Transform (FFT). Both branches are compressed to six dimensions and encoded in a six-qubit Variational Quantum Circuit (VQC) using orthogonal angle embedding and data re-uploading. On a balanced CIFAKE subset, the model achieves competitive predictive performance with only 72 trainable parameters in the quantum layer. An ablation study evaluates the contribution and computational cost of the main architectural components.

Keywords: Quantum Machine Learning; Transfer Learning; Variational Quantum Classifier; AI-Generated Image Detection.

Introduction



Figure 1: Pope Francis AI image wearing a jacket.

The rapid advancement of AI-based image generation models such as Flux2, GPT Image 1.5, and Nano Banana Pro has made synthetic images increasingly realistic.
While there is nothing inherently wrong with synthetic images, presenting an artificial image as real can have serious consequences for the information landscape.
The **malicious use** of artificial images can trigger waves of **fraud** across political, financial, and legal systems, as well as fuel the spread of **deepfakes** and non-consensual explicit imagery, creating a **crisis of structural trust**. Therefore, developing efficient methods to detect artificial images is crucial for preserving digital security.

Objectives

- Develop a dual-input model that fuses ResNet18 and FFT features
- Compresses the multimodal representation for a compact six-qubit VQC
- Compares the proposed architecture with classical and quantum baselines, and evaluates key ablations

Methodology

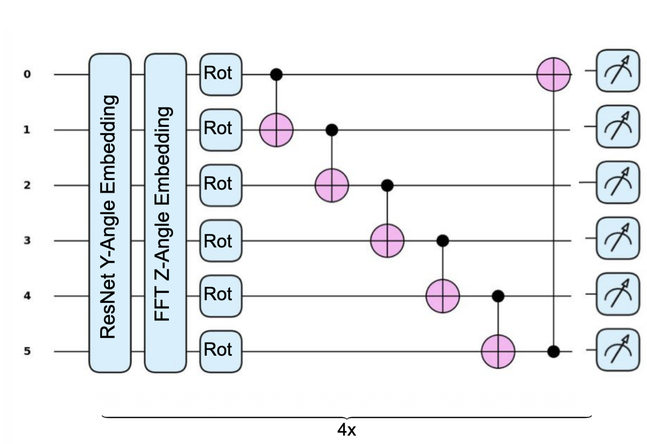


Figure 2: Pope Francis AI image wearing a jacket.

Dual-Input Feature Extraction and Compression. The proposed *Q-Trust AI* framework processes 32×32 input images through a dual-input classical pipeline to capture both visual semantics and frequency-domain artifacts:

- Visual Branch: A ResNet18 backbone (with its final residual block fine-tuned) extracts a 512-dimensional semantic feature vector. A classical dense layer compresses this vector to 6 dimensions.
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- Spectral Branch:A 2D Fast Fourier Transform (FFT) computes the radial power spectrum across 16 concentric frequency bins. A separate dense layer compresses this 16-dimensional vector to 6 dimensions.

Quantum Feature Encoding and VQC Architecture. The compressed 6-dimensional visual and spectral vectors are mapped into a 6-qubit Variational Quantum Circuit (VQC) via orthogonal Angle Embedding on the Bloch sphere:

- Orthogonal Embedding: Visual features parameterize R_Y rotations, while spectral features parameterize R_Z rotations, preventing destructive interference between domains.
- Circuit Design: The VQC comprises 4 data re-uploading layers. Each layer consists of parameterized single-qubit rotation gates $Rot(\theta, \phi, \lambda)$ (72 trainable quantum weights in total) followed by ring-connected CNOT gates to induce entanglement across neighboring qubits.

Readout, Softmax Classification, and Training Setup.

- Measurement and Softmax: The expectation values of the Pauli-Z observables $\langle Z_i \rangle$ ($i \in \{0, \ldots, 5\}$) are measured across all 6 qubits. These values pass through a classical linear layer, followed by a Softmax activation function to output normalized probability logits for the binary classification (P(Real) vs. P(Fake)).
- Dataset and Optimization: Evaluated on a balanced subset of $2,800$ samples ($2,000$ training / 400 validation / 400 test) from the *CIFAKE* benchmark with spatial data augmentation. Training was implemented in PyTorch and PennyLane using the AdamW optimizer, cosine annealing learning rate scheduler, dropout, gradient clipping, and label smoothing (95%).

Results

Accuracy: 90.75% · **F1: 0.9125** · **AUC: 0.962**
96% of synthetic images and **85%** of real images correctly identified.

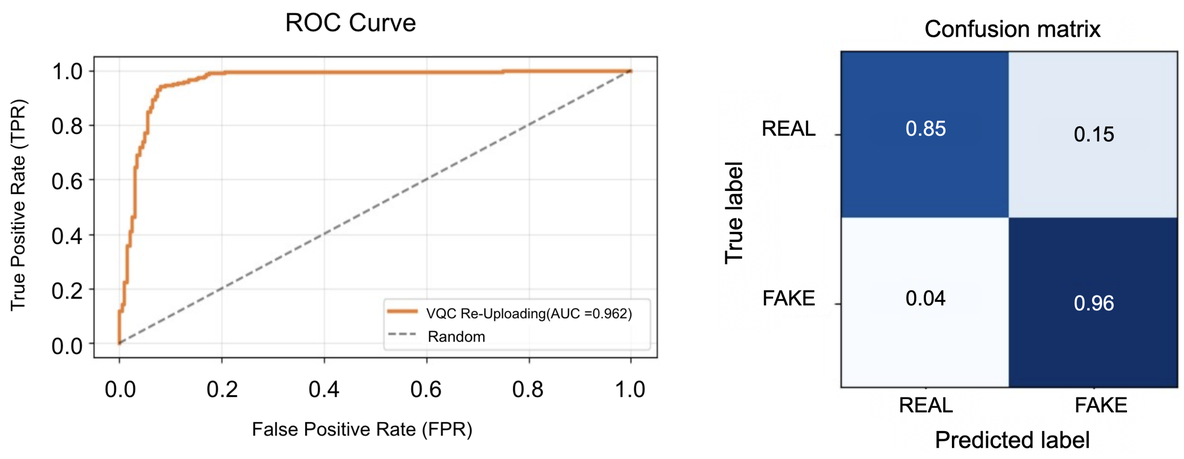


Figure 3: Performance metrics of ROC curve and normalized confusion matrix on the test set.

Key Takeaways

- Efficient hybrid quantum–classical architecture in the NISQ regime.
- Uses only 6 qubits.
- ResNet18 and FFT extract classical features.
- Data Re-Uploading improves circuit expressivity.
- The VQC captures non-linear correlations via entanglement.

Next Steps

- Test the model on Midjourney, Flux2, and other generators.
- Replace ring-CNOTs with Strongly Entangling Layers.
- Evaluate different numbers of layers.
- Explore other datasets and optimizers.
- Run the model on IBM Quantum or IonQ hardware.

References

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